

- 7 (a) When the switch is closed, current flows downwards in the wire. The magnetic field in the area ABCD interacts with the current, **a magnetic force** pointing to the **right** is created, by Fleming's Left Hand rule. B1

Force acts to **the right** and causes **an anticlockwise moment** about P, causing wire to be kicked out. B1

When the **wire leaves the mercury**, the closed circuit breaks, **no current flows**, the weight of the wire produces a **clockwise moment** to return the wire to the vertical position. B1

(b) Moment = $F \times d$

$$5.0 \times 10^{-4} = F \times (1.5 + 4.5) \times 10^{-2} \quad \text{B1}$$

$$F = 8.3 \times 10^{-3} \text{ N} \quad \text{A1}$$



[Turn over]

- (c) When the diameter is doubled, the resistance is $\frac{1}{4}$ of the original, so given the same voltage, current will be 4 times larger. Therefore the electromagnetic force will be 4 times larger. [1]

B1

Diameter doubles, the mass is 4 times; [1]

B1

Therefore, initial acceleration = $4F / 4M = F/M$.

the acceleration of the rod remains the unchanged. [1]

A1